

Security and Threats in Aviation: Cryptographic Based Network Security System

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Abstract—In present era, aviation communication technologies provides safe, effective, and efficient transfer of data between aircraft and facilities on the ground. Cryptography is essential to maintain the confidentiality, integrity, and legitimacy of data transfer in aviation systems. The safe key exchange and authentication that are essential components of aviation communication security are expertly ensured algorithms. Advanced Encryption Standard (AES), Rivest- Shamir- Adleman (RSA), Blowfish, Fernet, and Elliptic curve cryptog- raphy (ECC) are a few of the symmetric and asymmetric encryption method explored for aviation system. In this paper, the development and application of AES and Fernet encryption algorithms, specifically designed for safe client- server communication for the aviation industry sector, are the mainly implemented and performance analyzed. The public key is used for encryption, while the private key is used for decryption. The effectiveness of these algorithms in safeguarding data transfer between a client and server within an airline network is put to the test with great rigor.

Index Terms—Cryptography, Advanced Encryption Standard (AES), Rivest- Shamir-Adleman (RSA), Blowfish, Fernet, and Elliptic curve cryptography (ECC)

I. INTRODUCTION

Present day aircraft's incorporation of technology has molded the aviation sector by upgrading connectivity, efficiency, and safety. However, with this technical advance also comes a host of new security issues and dangers that could affect aviation systems. These systems' essential nature makes them an ideal target for hostile actors looking to take benefits of weaknesses for a number of reasons, from terrorism to espionage. The science of secure data transfer known as cryptography is important to protecting sensitive and valuable data in aircraft systems. It ensures authenticity, integrity and secrecy in communication channels by encrypting and decrypting data utilizing sophisticated algorithms. Cryptography is a crucial protective tactic against cyber-attacks since aviation systems rely heavily on data exchange between various on-board and ground-based systems.

Aircraft systems experience a variety of security issues even with the deployment of strong security measures. The vulnerability of the wireless communication networks within and beyond the airplane is one of the main worries. The increase in connectivity of aircraft to the netowrk either Internet or ground, increases the threat of unauthorized access and cyber-attacks. By encrypting critical data communication between aircraft components and preventing interception or manipulation by unauthorized parties, encryption methods in aviation systems reduce these threats. To counter advanced attack vectors, however, ongoing developments in cryptography techniques are required due to the ever- changing nature of cyber threats. The aviation security threat landscape is made up of a wide variety of possible dangers. The safety of flights is seriously threatened by cyber-attacks that target communication networks, navigational instruments, and flight control systems. Attacks by ransom ware, phishing, and malware on vital aviation infrastructure can result in data breaches, system failures, or even unapproved takeover of flight operations.

Advanced Encryption Standard (AES), Rivest- Shamir-Adleman (RSA), Blowfish, Fernet, and Elliptic curve cryptography (ECC) are just a few of the symmetric and asymmetric encryption methods that are explored. AES and Fernet encryption algorithms that are specially designed for safe client-server communication in the aviation industry are also being developed and tested as part of this study. Algorithms such as AES and Blowfish, which embody symmetric cryptography, utilize a solitary key for both encryption and decryption. These algorithms are highly efficient in managing substantial amounts of data, making them ideal for safe communication in aviation networks. A pair of keys is used in asymmetric cryptography, as demonstrated by algorithms like RSA and ECC. The public key is used for encryption, while the private key is used for decryption. The safe key exchange and authentication that are essential components of aviation communication security are expertly ensured by these algorithms.

In this paper, AES and Fernet encryption algorithms are the developed and implemented. These algorithms are specifically designed for the aviation communication sector which are the major work of this paper. The effectiveness of these algorithms in safeguarding data transfer between a client and server within an airline network is put to the test with great rigor.

The rest of the paper is arranged as follows: Section II presents work. Section III discusses the cryptography algorithms and results are analysed in Section IV. Finally Section V concludes the work.

II. RELATED WORK

A novel technology called AI Safe monitors communication between ATCO and aircrew in order to improve air traffic safety in the Air-Ground Collaboration (AGC), where speech communication is susceptible to Human-In-The-Loop (HITL) issues. The endmost objective is to increase operational safety and efficiency by automatically detecting threats[1].

The need for digital services is growing along with civil air traffic, which means that powerful and streamlined Air Traffic Management (ATM) systems are required. Therefore, it turns into imperative that ATM services be digitalized. Complete scientific studies describing security breaches across applications, networks, and data connection layers in civil aviation communication systems are presently lacking. Cost reduction in aviation is still a major concern, notwithstanding the significance of cybersecurity. Aeronautical systems locate it tough to update security measures since recertification is required for even small changes, which include upgrades to cryptographic algorithms[2]. Non-important aviation communications along with route modifications and level assignments, are catered to by CPDLC. Notwithstanding its advantages, CPDLC's weak cybersecurity safeguards put the system as a entire at danger. When messages are despatched across unprotected wireless networks in unencrypted, hackers can take advantage of them. For example, an attacker can regulate messages and snoop on CPDLC traffic using very little hardware[3].

Automatic Dependent Surveillance-Broadcast (ADS-B) depends on airplanes often transmitting encoded data, which anybody with the proper device can intercept and decode. ADS-B is susceptible to spoofing techniques such as Flight Data Injection techniques (FDIA) due to its openness. Unsupervised ML models called autoencoders are frequently employed in anomaly detection applications. In order to recreate the original data, they first encode the input data into a lower-dimensional representation, which they then decode. An asymmetric autoencoder called Contextual Auto-Encoder (CAE) was created expressly to locate abnormalities in time-series data. The method addresses contextual fluctuations in time series by use of a single encoder that is coupled to many decoders[4].

Data is used by AI and deep learning to create new rules, forecast results, and find trends and anomalies. In order to decrease accidents in transportation systems, safety concerns have prompted researchers to concentrate on intelligent systems, behaviour analysis, accident modelling, and object

identification. Studies suggest that AI has the potential to improve aviation trip management. Abduljabbar et al. (2019) employed outlier analysis on 2,200 flights and unsupervised machine learning to cluster landing phase-sequences. Their strategy seems to have the potential to improve landing phase safety[5].

Threats from terrorism have affected civil aviation for more than 50 years, which has resulted in ongoing improvements to aviation security. Every security incident led to the discovery of flaws and modifications to the current protocols. Proactive security relies heavily on unpredictability. The goal is to maximize the effectiveness and deterrent effect of measures such as diversified patrol operations, identity checks, quota-based screenings, worker rotations, and random security checks on goods and passengers[6].

A commercial aviation system is represented by a Transport Class Model (TCM) that incorporates the Model Reference Adaptive Control (MRAC), a sophisticated intelligent controller that synchronizes the aircraft's output with a reference model. When an FDI attack is introduced into pilot commands (such as elevator commands), the AIS algorithm is triggered to detect and eliminate malicious data packets. The NARX network is then used to forecast and make up for dropped packets[7].

Communication, Navigation, and Surveillance (CNS) systems are essential to the aviation ecosystem, which includes Air Traffic Management (ATM). Information sharing between equipment, systems, and users—like pilots and air traffic control (ATC)—is made possible via communication systems. Nevertheless, these connections are vulnerable to malicious attacks due to their wireless nature. Attacks pertaining to navigation include a wide range of risks, including signal jamming, eavesdropping, GPS spoofing or blocking, single-tone frequency manipulation, and data alterations. Unauthorized surveillance of aircraft movements is the goal of surveillance-related attacks, which use techniques including signal jamming, modification, or deletion. Within the domain of aircraft communication and navigation, these attacks present serious hazards[8].

To evaluate the risks associated with enterprise security, it seems that a threat analysis ontology derived from NIST was used as the basis for the security study. Entities such as threat actors, vulnerabilities, operational impact, e-Enabled domains, target assets, aviation tactics, attack methodologies, and procedures are all included in the ontology. Cyberattacks on the ADS-B System: Basic security features like as encryption, message integrity, and authentication are absent from the ADS-B system. These security flaws make it dangerous for airplanes to provide real-time information in crowded airspace, leaving them open to many types of attacks[9].

Symmetric and asymmetric cryptosystems are the two basic categories into which encryption techniques often fall. A public-private key pair is used in asymmetric cryptosystems, whereas the same key is used in symmetric cryptosystems for both encryption and decryption. The communication is encrypted by the sender using the recipient's public key, and

it is decrypted by the recipient using their private key. An asymmetric encryption technique known as the RSA algorithm provides features such as robustness against different types of attacks, security based on prime number size (e.g., a 1024-bit key is almost secure), and the impossibility to extract the private key from the public key. However, the recovered cipher image may have deformation during encryption and decryption processes[10].

Device connections over the internet are made easier by IoT. Sensors are arranged for wireless communication, information processing, and detection using wireless sensor networks, or WSNs. WSN security is improved by RSA cryptography, which protects data integrity and transmission. Encryption and decryption of shared data are made possible by Key Generation, which is essential to estimating Trust Value[11-17].

III. CRYPTOGRAPHIC ALGORITHMS

Cryptography is technique of securing information and communications through use of codes so that only those people for whom the information is intended can understand it and process it. Thus preventing unauthorized access to information. The prefix “crypt” means “hidden” and suffix “graphy” means “writing”. In Cryptography, the techniques that are used to protect information are obtained from mathematical concepts and a set of rule- based calculations known as algorithms to convert messages in ways that make it hard to decode it. These algorithms are used for cryptographic key generation, digital signing, and verification to protect data privacy, and web browsing on the internet and to protect and send confidential information.

A. Elliptical curve cryptography (ECC)

The algebraic structure of elliptic curves over finite fields is used by the Elliptic Curve Cryptography (ECC) cryptographic technique to enable safe communication. It is predicated on how hard the Elliptic Curve Discrete Logarithm Problem (ECDLP) is. In this problem, k is a number that must be found in the formula $k = P + kG$, where P is a point on the curve and G is the generating point.

Advantages:

- Strong Security: Compared to other algorithms like RSA, it offers a higher level of security with lower key lengths, resulting in a more efficient use of computational resources.
- Efficiency: ECC works well in contexts with limited resources, such as mobile and Internet of Things devices, because it takes less memory and processing power.

Disadvantages:

- Complexity: In order to prevent weaknesses, the implementation of ECC might be complicated and need careful handling.
- Patent Issues: There may have been patent-related problems with ECC implementations in the past, although many of these have already expired or lost significance.

```
PS C:\Users\asdfs> & C:\Users\asdfs\AppData\Local\Microsoft\WindowsApps\python3.11.exe c:/Users/asdfs/OneDrive/Desktop/ha
sh/CODES/verified/ctr3.py
AES ENCRYPT KEY: b'ab8ca6ef702c72f0fe3542d7886d42f0'
ENTER THE MESSAGE:Hello World!
Encrypted msg: b'ecf962c4a8398ea2ea0406cf'
DECRYPTED TEXT: Hello World!
```

Fig. 1. AEC Encryption

```
PS C:\Users\asdfs> & C:\Users\asdfs\AppData\Local\Microsoft\WindowsApps\python3.11.exe c:/Users/asdfs/OneDrive/Desktop/ha
sh/CODES/verified/rsat.py
ENTER THE MESSAGE:Hello World!
original Message: Hello World!
Encrypted msg: b'\xafaf\ybc\x14tj]\x0et\x27\xab\x23\xed\xab~\x2d\vf\x01\x13\\x0c\x14\xea\x23\x27)\xfdf\x1f'
decrypted: Hello World!
```

Fig. 2. RSA Encryption

B. Advanced Encryption Standard (AES)

Sensitive data is protected using a symmetric encryption algorithm called the Advanced Encryption Standard (AES). It uses cryptographic keys of different lengths (128, 192, or 256 bits) to work on fixed-size data blocks.

Advantages:

- Security: AES is a dependable option for encryption because it is thought to be extremely secure and has withstood thorough cryptanalysis.
- Efficiency: This method is ideal for a variety of applications due to its reasonably fast execution time and efficiency in both software and hardware implementations.

Disadvantages:

- Key management: It's important to manage keys properly because improper handling or compromise of keys can jeopardize AES security.
- Resource-intensive: Although AES encryption and decryption are efficient, they can nevertheless demand a substantial amount of computer power, particularly when dealing with big data sets.

C. Rivest- Shamir-Adleman (RSA)

A public-key cryptography technique called the RSA algorithm is utilized for digital signatures and safe data transmission. It is predicated on how hard it is to calculate huge prime numbers mathematically.

Advantages:

- Security: Because factoring big numbers is difficult, RSA offers strong security by making it difficult for unauthorized parties to decipher messages without the private key.
- Digital Signatures: RSA makes it possible to create safe digital signatures that confirm the legitimacy of communications and guarantee they haven't been tampered with.

Disadvantages:

- Performance: In contexts with limited resources, the computational demands of RSA can be significant, particularly when dealing with bigger key sizes.
- Key management: Keeping track of and safely storing keys can be difficult, particularly when working with large-scale systems and numerous keys that have distinct uses.

```
In [33]: runfile('C:/Users/91866/Desktop/harsha/hash/hp/hash/CODES/verified/crypt_fernet.py', wdir='C:/Users/91866/Desktop/harsha/hash/hp/hash/CODES/verified')

ENTER THE MESSAGE:Hello World!
original msg: Hello World!
encrypted:
b'gAAAAABlwYwSeAlwERkRx1c07C2iq9Jeo3ta4f27_XpHGrAHe8WeRF70twbIKj7eGJs-0WH8LRRLiLSbgoaXTRCcfMn4PhbzA=='
decoded msg: Hello World!
```

Fig. 3. Fernet Encryption

```
In [32]: runfile('C:/Users/91866/Desktop/harsha/hash/hp/hash/CODES/verified/blowfish.py', wdir='C:/Users/91866/Desktop/harsha/hash/hp/hash/CODES/verified')

Enter data to encrypt: 562490
Encrypted data : 12536057666182640328
Decrypted data : 562490
```

Fig. 4. Blowfish Encryption

D. Fernet Cryptographic

A symmetric encryption technique for safe communication, the Fernet algorithm is frequently implemented in Python's cryptography module.

Advantages:

- Security: To ensure data confidentiality and integrity, it uses robust encryption methods including AES-128 in CBC mode and HMAC for message authentication.
- Easy to Use: Fernet offers a basic encryption and decryption interface, making secure communication implementation simple for developers.

Disadvantages:

- Key management can be difficult, particularly in distributed systems, since Fernet is a symmetric encryption technique that calls for safe key exchange or storage.
- Limited Functionality: Fernet may not be appropriate for all encryption use cases, such as asymmetric encryption for key exchange, even though it performs a great job at encrypting data in transit or at rest.

E. Blowfish Algorithm

Using a variable-length key, the Blowfish algorithm is a symmetric-key block cipher that encrypts data in fixed-size blocks.

Advantages:

- Speed: When compared to certain other encryption techniques, Blowfish is rather quick.
- Flexibility: It may be adjusted to meet various security requirements because it offers varied key lengths.

Disadvantages:

- Security Concerns: As cryptanalysis advances, more vulnerabilities can appear over time.
- Restricted Block Size: In some applications, when higher block sizes are required, the fixed block size may make it less useful.

```
In [5]: runfile('C:/Users/ASDG/Desktop/hash/getac/s.py', wdir='C:/Users/ASDG/Desktop/hash/getac')
b'YLyH8hG3uZP92d_R62tAZhamQ13_daGrHYFsFNTjJrW='
The time taken to encrypt and send the data: 61.96188926696777 ms
```

Fig. 5. Runfile for transmission using Fernet Algorithm

```
In [9]: runfile('C:/Users/91866/Desktop/harsha/hash/getac/hash/getac/r.py', wdir='C:/Users/91866/Desktop/harsha/hash/getac/hash/getac')
172.20.10.4
The time taken to receive the encrypted file and decrypt it in another text file: 7688.625812530518 ms
```

Fig. 6. Runfile for decryption using Fernet Algorithm

IV. RESULT ANALYSIS

Two algorithms; Fernet and AEC are mainly implemented and analysed for Cryptographic Network system. The Fig. 5-8 represents the runfile for encryption and decryption process using Fernet and AES algorithms.

A. Cryptographic Network system (Fernet - Client to Server)

A basic file encryption process using the Fernet encryption module from the cryptography library in Python. The process involves:

Encryption (Server Side)

- Reads the content of the file ('i.txt') to be encrypted.
- Generates a random key using Fernet.
- Encrypts the file data using the generated key.
- Saves the encrypted content into a new file ('encrypted.txt').
- Initiates a socket connection to send the encrypted file and key.

Transmission:

- Establishes a socket connection to the receiver's IP and port.
- Sends the encrypted file content and the encryption key over the established connection.

Decryption (Client Side):

- The client receives the encrypted file and the encrypted key through the established connection.
- The client performs decryption using the received key to retrieve the original file content.

B. Cryptographic Network system (AES - Client to Server)

A secure file transmission mechanism between two devices (sender and receiver) using the Advanced Encryption Standard (AES) algorithm for encryption and socket communication for data exchange.

Procedures for Sender Code (Device 1):

Derivation and Encryption of Keys

- PBKDF2HMAC generates cryptographic keys using a password, guaranteeing a secure key generation procedure.

```
In [6]: runfile('C:/Users/ASDG/Desktop/hash/getac/s4.py',
wdir='C:/Users/ASDG/Desktop/hash/getac')
The time taken to encrypt and send: 24.98551834106445 ms
```

Fig. 7. Runfile for encryption using AES algorithm

```
In [10]: runfile('C:/Users/91866/Desktop/harsha/hash/getac/hash/getac/
r4.py', wdir='C:/Users/91866/Desktop/harsha/hash/getac/hash/getac')
connected by ('172.20.10.14', 62613)
The time taken for receive the encrypted file and decrypt the file is
5680.502414703369 ms
```

Fig. 8. Runfile for decryption using AES algorithm

- AES encryption is applied to the file that needs to be sent.
- To ensure confidentiality, the file content is encrypted using AES in Cipher Feedback (CFB) mode.

Communication Socket:

- To connect the recipient device to the network, a socket is made.
- Over the established socket connection, the encrypted file content, initialization vector (IV), and derived salt are delivered.

Procedures for Receiver Code (Device 2): Establishing and Connecting the Socket

- On a designated port, a socket is configured to wait for incoming connections.
- The recipient confirms the connection after it has been made by the sender.

Acquiring and Interpreting:

- Through the socket connection, the recipient receives the transmitted IV, salt, and encrypted file content.
- The recipient determines the decryption key by using the same password and the received salt.
- Using the derived key and IV, AES decryption is applied to the received encrypted content to recover the original file content.

V. CONCLUSION

Cryptographic based network security system is analysed for security and threats issues in aviation. Two algorithms; AES and Fernet encryption algorithms are implemented and analysed. The encryption and decryption of network security system is studied using AES and Fernet algorithms. The future of security and threats in aviation may be shaped via a dynamic interaction of technological advancements, evolving threat landscapes, and regulatory frameworks. Continuous innovation, collaboration, and version may be critical in addressing those demanding situations and making sure the safety and security of the aviation industry.

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